End-to-end, decision-based, cardinality-constrained portfolio optimization

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Outline

Motivation & Background

Predict-then-optimize v.s. Decision-focused learning

Given the covariance of N assets' returns $\Sigma \in \mathbb{R}^{N \times N}$, an investor often needs to solve a sparse portfolio optimization problem:

$$\min_{\mathbf{w} \in \mathcal{W}_k} \mathbf{w}^\top \Sigma \mathbf{w}, \qquad \mathcal{W}_k = \{\mathbf{w} \in \mathbb{R}_+^N : \mathbf{1}^\top \mathbf{w} = 1, \|\mathbf{w}\|_0 \le k\}. \tag{P}$$

Predict, then optimize

- Decoupled. First use traditional statistical methods to estimate Σ, then solve the optimization problem.
- Trained for accuracy, not decision quality.

Decision-focused learning

- End-to-end. Train the model parameters through the downstream optimization problem.
- E.g., Smart "predict, then optimize" [1].

Cardinality constraints Portfolio optimization

There are several ways of formulation to the problem

$$\mathcal{W}_k = \{\mathbf{w} \in \mathbb{R}_+^N : \mathbf{1}^\top \mathbf{w} = 1, \|\mathbf{w}\|_0 \leq k\}.$$

By introducing binary indicators $z_i = \mathbb{I}\{w_i > 0\}$ to indicate if asset i is included in the portfolio, the problem can be reformulated as a mixed-integer program (MIP):

- *Big-M formulation*:

$$\mathcal{W}_k^M = \left\{\mathbf{w} \in \mathbb{R}_+^N,\ \mathbf{z} \in \{0,1\}^N : \mathbf{1}^\top \mathbf{w} = 1,\ \mathbf{1}^\top \mathbf{z} = k,\ w_i \leq z_i,\ i = 1, \ldots, N\right\}.$$

- *Complementarity-type formulation* (tighter):

$$\mathcal{W}_k^C = \left\{\mathbf{w} \in \mathbb{R}_+^N,\ \mathbf{z}^c \in \{0,1\}^N : \mathbf{1}^\top \mathbf{w} = 1,\ \mathbf{1}^\top \mathbf{z}^c = k,\ w_i z_i^c = 0,\ i = 1, \ldots, N\right\}.$$

Factor Models in Finance i

- Asset returns are modeled as **common factors** plus **idiosyncratic noise**:

$$r_{it} = \alpha_i + \sum_{p=1}^{P} \beta_{ip} f_{pt} + \varepsilon_{it} \quad \Longleftrightarrow \quad \mathbf{R} = \mathbf{1}\boldsymbol{\alpha}^\top + \mathbf{F}\mathbf{B}^\top + \boldsymbol{\varepsilon}.$$

  Here $\mathbf{R} \in \mathbb{R}^{T\times N}$, $\mathbf{F} \in \mathbb{R}^{T\times P}$, $\boldsymbol{\alpha} \in \mathbb{R}^{N}$, $\mathbf{B} \in \mathbb{R}^{N\times P}$, and $\boldsymbol{\varepsilon} \in \mathbb{R}^{T\times N}$.
- **Fama-French five-factor** model [2]: `Mkt-Rf` (market risk premium), **SMB** (size), **HML** (value), **RMW** (profitability), and **CMA** (investment style).
- Factors are *known, observable (at the end of the trade-day), and shared by all assets*; factor loadings β_{ip} and idiosyncratic errors ε_{it} are asset-specific:

$$\mathbf{F} = \begin{bmatrix} \texttt{Mkt-RF}_1 & \texttt{SMB}_1 & \texttt{HML}_1 & \texttt{RMW}_1 & \texttt{CMA}_1 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \texttt{Mkt-RF}_T & \texttt{SMB}_T & \texttt{HML}_T & \texttt{RMW}_T & \texttt{CMA}_T \end{bmatrix} \in \mathbb{R}^{T\times 5}.$$

Factor Models in Finance ii

By denoting $\mathrm{Cov}(\mathbf{F}) := \Sigma_f \in \mathbb{R}^{P\times P}$, $\mathrm{Cov}(\boldsymbol{\varepsilon}) := \boldsymbol{\Psi} \in \mathbb{R}^{N\times N}$ (often assumed to be diagonal), and the asset covariance as $\Sigma \in \mathbb{R}^{N\times N}$, and taking the covariance of the return model $\mathbf{R} = \mathbf{1}\boldsymbol{\alpha}^\top + \mathbf{F}\mathbf{B}^\top + \boldsymbol{\varepsilon}$ implies

$$\Sigma = \mathbf{B}\Sigma_f\mathbf{B}^\top + \boldsymbol{\Psi} = \mathbf{B}\Sigma_f\mathbf{B}^\top + \mathrm{diag}(\psi_1^2, \ldots, \psi_N^2).$$

Remark

- The factor covariance Σ_f can be directly computed from the data.
- The parameters to be estimated are the factor loadings $\mathbf{B}$ and the idiosyncratic variances $\boldsymbol{\Psi}$.
- For convenience, one may collect the parameters into a single vector $\boldsymbol{\theta} = (\mathbf{B}, \boldsymbol{\Psi})$.

Methodology

Notation reference

Symbol	Meaning
N, k	number of assets, target cardinality
$j \in \{1, \ldots, J\}$	problem instance index (for Bootstrap indexing)
$\mathbf{w} \in \mathbb{R}_+^N$	long-only portfolio weights with $\mathbf{1}^\top \mathbf{w} = 1$
$\mathbf{z} \in \{0,1\}^N, \tilde{\mathbf{z}} \in [0,1]^N$	asset-selection indicators, continuous relaxation of $\mathbf{z}$
$\mathbf{F}^{(j)} \in \mathbb{R}^{T \times P}$, $\mathbf{R}^{(j)} \in \mathbb{R}^{T \times N}$	factor returns, asset returns for problem instance j
$\mathbf{B} \in \mathbb{R}^{N \times P}$	factor loading matrix
$\mathbf{\Psi} = \mathrm{diag}(\psi_1^2, \ldots, \psi_N^2)$	diagonal idiosyncratic covariance
$\Sigma_f^{(j)} \in \mathbb{R}^{P \times P}$	sample covariance of factors in instance j
$\Sigma_\theta^{(j)} \in \mathbb{R}^{N \times N}$	model-implied asset covariance matrix (Σ if not estimated by factor model)
ℓ_d	decision-based realized loss

Methodology

Predict, then optimize baseline

LinReg: Decoupled baseline (predict, then optimize)

Input: Factors $\mathbf{F}^{(j)}$, returns $\mathbf{R}^{(j)}$, cardinality k

1 Fitting the factor model $\mathbf{R} = \mathbf{1}\boldsymbol{\alpha}^\top + \mathbf{F}\mathbf{B}^\top + \boldsymbol{\varepsilon}$ directly on accuracy accuracy loss ℓ_a to obtain $\hat{\boldsymbol{\theta}}^{(j)} = (\hat{\mathbf{B}}^{(j)}, \hat{\boldsymbol{\Psi}}^{(j)})$.

2 Construct the covariance $\Sigma_{\hat{\theta}}^{(j)} = \hat{\mathbf{B}}^{(j)} \Sigma_f^{(j)} \hat{\mathbf{B}}^{(j)\top} + \hat{\boldsymbol{\Psi}}^{(j)}$.

3 Solve the cardinality-constrained problem with by Gurobi to obtain $\mathbf{w}^{(j)} := \arg\min_{w \in \mathcal{W}_k} \mathbf{w}^\top \Sigma_{\hat{\theta}}^{(j)} \mathbf{w}$.

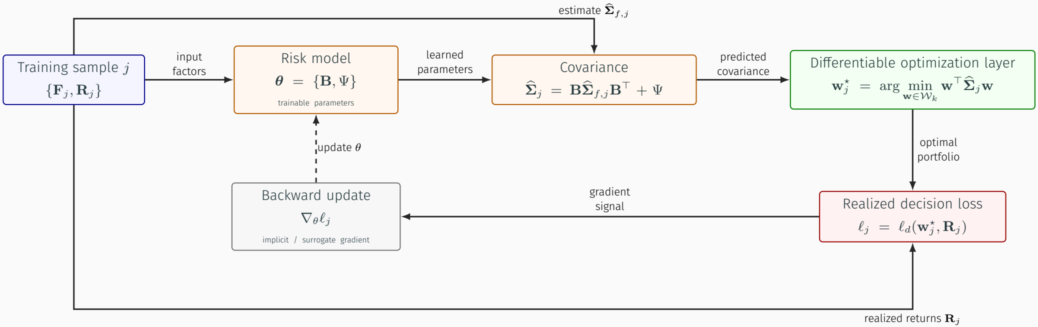
Output: Portfolio weights $\mathbf{w}^{(j)}$

- Accuracy loss does not see the sparse optimizer.
- The fitted covariance is treated as deterministic.
- Mean-variance portfolios are highly sensitive to estimation error.

Methodology

End-to-end framework

End-to-End architecture as a computational graph



Limitation of current MIP formulation & relaxation layers

The dependency $\ell_d\left(\arg\min_{\mathbf{w}\in\mathcal{W}_k} \mathbf{w}^\top \Sigma_\theta \mathbf{w}, \mathbf{R}\right)$ creates two obstacles:

1. $\mathcal{W}_k$ is discrete; the optimizer jumps between supports.
2. Even for continuous problems, differentiating $\arg\min$ requires sensitivity analysis.

Thus, consider the following three possible continuous convex relaxations to remove discreteness:

Layer	Main tradeoff
E2E M	Loosest but computationally cheapest
E2E SOCP [3]	Tighter; conic constraints add cost
E2E SDP [4]	Tightest target; much larger decision layer

For instance j, the Big-M relaxation layer solves

$$\min_{\mathbf{w},\tilde{\mathbf{z}}} \quad \mathbf{w}^\top \mathbf{B}\Sigma_f^{(j)}\mathbf{B}^\top \mathbf{w} + \mathbf{w}^\top \mathbf{\Psi} \mathbf{w}$$

subject to

$$\mathbf{1}^\top \mathbf{w} = 1, \qquad \mathbf{1}^\top \tilde{\mathbf{z}} \leq k, \qquad \mathbf{w} \leq \tilde{\mathbf{z}},$$
$$0 \leq \tilde{\mathbf{z}} \leq \mathbf{1}, \qquad \mathbf{w} \geq 0,$$

Remark

- $\tilde{\mathbf{z}} \in [0,1]^N$ is the continuous relaxation of indicators $\mathbf{z} \in \{0,1\}^N$.
- Note that $\mathbf{1}^\top \tilde{\mathbf{z}} \leq k$ is now a soft constraint, so the solution may have more than k nonzeros.
- Empirically, this loose layer is the fastest and often performs best.

E2E Big-M relaxation ii

Moreover, note that the idiosyncratic risk term $\mathbf{\Psi}$ is a diagonal matrix with positive entries. Thus, to keep this specific structure during gradient updates, the actual updated parameter is

$$\hat{\boldsymbol{\psi}} = [\hat{\psi}_1, \dots, \hat{\psi}_N]^\top,$$

and the final idiosyncratic covariance is given by

$$\mathbf{\Psi} = \operatorname{diag}\left(\psi_1^2, \dots, \psi_N^2\right) := \begin{pmatrix} \exp(\hat{\psi}_1)^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \exp(\hat{\psi}_N)^2 \end{pmatrix}.$$

This is also the case for the remaining two relaxations.

E2E SOCP relaxation

The SOCP relaxation separates systematic and idiosyncratic risk:

$$\min_{\mathbf{w},\tilde{\mathbf{z}},\boldsymbol{\delta}} \mathbf{w}^\top \mathbf{B}\Sigma_f \mathbf{B}^\top \mathbf{w} + \mathrm{diag}(\Psi)^\top \boldsymbol{\delta}$$

subject to

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{1}^\top \tilde{\mathbf{z}} \leq k, \quad \mathbf{w} \leq \tilde{\mathbf{z}}, \quad w_i^2 \leq \delta_i \tilde{z}_i \ (\forall i), \quad 0 \leq \tilde{\mathbf{z}} \leq \mathbf{1}, \quad \mathbf{w} \geq 0, \quad \boldsymbol{\delta} \geq 0.$$

Intuition of δ_i

- Consider a toy model: $\min_w \psi^2 w^2$, s.t., $\psi^2 > 0$. By introducing δ and get $\min_{w,\delta} \psi^2 \delta$, s.t., $w^2 \leq \delta$. The optimal solution is $\delta^\star = w^{\star 2}$, and the objective value is $\psi^2 w^{\star 2}$, which is the same as the original problem.
- Then consider $w_i^2 \leq \delta_i z_i$ with $z_i \in \{0, 1\}$. If $z_i = 1$, then $w_i^2 \leq \delta_i$ and it is the same as the toy model. If $z_i = 0$, then $w_i^2 \leq 0$ implies $w_i = 0$, which is consistent with the original problem.
- For relaxed term when $\tilde{z}_i \in (0, 1)$, the constraint $\delta_i \geq w_i^2 / \tilde{z}_i > w_i^2$. It imposes a stronger penalty on δ_i than the original problem, which encourages $\tilde{z}_i$ to be close to 0 or 1, and discourages fractional solutions (hesitant solutions).

E2E SDP relaxation layer

The SDP reformulation is

$$\min_{\mathbf{U}} \ \langle \Sigma_\theta, \mathbf{W} \rangle$$

subject to

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{1}^\top \tilde{\mathbf{z}}^c = N - k, \quad \mathbf{w}, \tilde{\mathbf{z}}^c \geq 0, \quad \mathrm{diag}(\tilde{\mathbf{Z}}^c) = \tilde{\mathbf{z}}^c, \quad \mathrm{diag}(\mathbf{Q}) = 0, \quad \mathbf{U} = \begin{pmatrix} 1 & \mathbf{w}^\top & \tilde{\mathbf{z}}^{c\top} \\ \mathbf{w} & \mathbf{W} & \mathbf{Q} \\ \tilde{\mathbf{z}}^c & \mathbf{Q}^\top & \tilde{\mathbf{Z}}^c \end{pmatrix} \succeq 0.$$

(Very naive) Intuition of $\mathbf{U} \succeq 0$

- According to the Schur complement condition, $\mathbf{U} \succeq 0$ is equivalent to $\begin{pmatrix} \mathbf{W} - \mathbf{w}\mathbf{w}^\top & \mathbf{Q} - \mathbf{w}\tilde{\mathbf{z}}^{c\top} \\ \mathbf{Q}^\top - \tilde{\mathbf{z}}^c\mathbf{w}^\top & \tilde{\mathbf{Z}}^c - \tilde{\mathbf{z}}^c\tilde{\mathbf{z}}^{c\top} \end{pmatrix} \succeq 0$. This means that the lifted variables $\mathbf{W}$, $\tilde{\mathbf{Z}}^c$, and $\mathbf{Q}$ are consistent with the original variables $\mathbf{w}$ and $\tilde{\mathbf{z}}^c$ in a positive semidefinite sense.
- Or equivalently, for exact lifted solutions, one hopes $\mathbf{U}_{\text{exact}} := \begin{pmatrix} 1 & \mathbf{w}^\top & \tilde{\mathbf{z}}^{c\top} \\ \mathbf{w} & \mathbf{W} & \mathbf{Q} \\ \tilde{\mathbf{z}}^c & \mathbf{Q}^\top & \tilde{\mathbf{Z}}^c \end{pmatrix} = \begin{pmatrix} 1 & \mathbf{w}^\top & \tilde{\mathbf{z}}^{c\top} \\ \mathbf{w} & \mathbf{w}\mathbf{w}^\top & \mathbf{w}\tilde{\mathbf{z}}^{c\top} \\ \tilde{\mathbf{z}}^c & \tilde{\mathbf{z}}^c\mathbf{w}^\top & \tilde{\mathbf{z}}^c\tilde{\mathbf{z}}^{c\top} \end{pmatrix}$, which mathematically implies $\mathbf{U}_{\text{exact}} \succeq 0$ and $\mathbf{rank}(\mathbf{U}_{\text{exact}}) = 1$. The SDP relaxation drops the rank-1 constraint (as it is non-convex), but the PSD constraint still encourages low-rank solutions.
- To conclude with, the PSD constraint $\mathbf{U} \succeq 0$ implies that $\mathbf{W} \approx \mathbf{w}\mathbf{w}^\top$, $\tilde{\mathbf{Z}}^c \approx \tilde{\mathbf{z}}^c\tilde{\mathbf{z}}^{c\top}$, and $\mathbf{Q} \approx \mathbf{w}\tilde{\mathbf{z}}^{c\top}$, which is the key to the tightness of the SDP relaxation.

E2E SDP relaxation layer

The SDP reformulation is

$$\min_{\mathbf{U}} \ \langle \Sigma_\theta, \mathbf{W} \rangle$$

subject to

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{1}^\top \tilde{\mathbf{z}}^c = N - k, \quad \mathbf{w}, \tilde{\mathbf{z}}^c \geq 0, \quad \mathrm{diag}(\tilde{\mathbf{Z}}^c) = \tilde{\mathbf{z}}^c, \quad \mathrm{diag}(\mathbf{Q}) = 0, \quad \mathbf{U} = \begin{pmatrix} 1 & \mathbf{w}^\top & \tilde{\mathbf{z}}^{c\top} \\ \mathbf{w} & \mathbf{W} & \mathbf{Q} \\ \tilde{\mathbf{z}}^c & \mathbf{Q}^\top & \tilde{\mathbf{Z}}^c \end{pmatrix} \succeq 0.$$

Intuition of the objective

- Still consider the exact term $\mathbf{W} = \mathbf{w}\mathbf{w}^\top$. Then the original objective $\mathbf{w}^\top \Sigma_\theta \mathbf{w} = \mathrm{Tr}(\Sigma_\theta \mathbf{w}\mathbf{w}^\top) = \langle \Sigma_\theta, \mathbf{W} \rangle$.
- Thus, current version is an approximate lifted surrogate for $\mathbf{w}\mathbf{w}^\top$.

E2E SDP relaxation layer

The SDP reformulation is

$$\min_{\mathbf{U}} \ \langle \Sigma_\theta, \mathbf{W} \rangle$$

subject to

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{1}^\top \tilde{\mathbf{z}}^c = N - k, \quad \mathbf{w}, \tilde{\mathbf{z}}^c \geq 0, \quad \mathrm{diag}(\tilde{\mathbf{Z}}^c) = \tilde{\mathbf{z}}^c, \quad \mathrm{diag}(\mathbf{Q}) = 0, \quad \mathbf{U} = \begin{pmatrix} 1 & \mathbf{w}^\top & \tilde{\mathbf{z}}^{c\top} \\ \mathbf{w} & \mathbf{W} & \mathbf{Q} \\ \tilde{\mathbf{z}}^c & \mathbf{Q}^\top & \tilde{\mathbf{Z}}^c \end{pmatrix} \succeq 0.$$

Intuition of the constraints w.r.t. $\mathbf{Z}^c$

- For exact lifted solutions with binary indicators $\mathbf{Z}^c_{\text{exact}} = \mathbf{z}^c \mathbf{z}^{c\top}$, its diagonal entries are $(\mathbf{Z}^c_{\text{exact}})_{ii} = z_i^c z_i^c = z_i^c$, i.e. $\mathrm{diag}(\mathbf{Z}^c_{\text{exact}}) = \mathbf{z}^c$. In the relaxed case, we keep this necessary condition $\mathrm{diag}(\tilde{\mathbf{Z}}^c) = \tilde{\mathbf{z}}^c$, without enforcing the full rank-1 constraint $\tilde{\mathbf{Z}}^c = \tilde{\mathbf{z}}^c \tilde{\mathbf{z}}^{c\top}$.
- Furthermore, $\mathbf{U} \succeq 0$ implies $\tilde{\mathbf{Z}}^c - \tilde{\mathbf{z}}^c \tilde{\mathbf{z}}^{c\top} \succeq 0$, which means $(\tilde{\mathbf{Z}}^c)_{ii} \geq (\tilde{\mathbf{z}}^c)_i^2$ for all i. Given that $(\tilde{\mathbf{Z}}^c)_{ii} = (\tilde{\mathbf{z}}^c)_i$, we have $(\tilde{\mathbf{z}}^c)_i \geq (\tilde{\mathbf{z}}^c)_i^2$, which implies $(\tilde{\mathbf{z}}^c)_i \in [0, 1]$ for all i. Thus, the box constraint $0 \leq \tilde{\mathbf{z}}^c \leq \mathbf{1}$ is inherent in $\mathbf{U} \succeq 0$.

E2E SDP relaxation layer

The SDP reformulation is

$$\min_{\mathbf{U}} \langle \Sigma_\theta, \mathbf{W} \rangle$$

subject to

$$\mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{1}^\top \tilde{\mathbf{z}}^c = N - k, \quad \mathbf{w}, \tilde{\mathbf{z}}^c \geq 0, \quad \operatorname{diag}(\tilde{\mathbf{Z}}^c) = \tilde{\mathbf{z}}^c, \quad \operatorname{diag}(\mathbf{Q}) = 0, \quad \mathbf{U} = \begin{pmatrix} 1 & \mathbf{w}^\top & \tilde{\mathbf{z}}^{c\top} \\ \mathbf{w} & \mathbf{W} & \mathbf{Q} \\ \tilde{\mathbf{z}}^c & \mathbf{Q}^\top & \tilde{\mathbf{Z}}^c \end{pmatrix} \succeq 0.$$

Intuition of the constraints w.r.t. $\mathbf{Q}$

- Similarly, for binary indicators, the original complementarity constraint is $w_i z_i^c = 0$ for all i. Thus in the exact lifted case, we have $\mathbf{Q}_{\text{exact}} = \mathbf{w}\tilde{\mathbf{z}}^{c\top}$, and $w_i z_i^c = 0$ implies $\operatorname{diag}(\mathbf{Q}_{\text{exact}}) = \mathbf{0}$.
- In the relaxed case, the PSD condition will gives $(\mathbf{W}_i i - w_i^2)\tilde{z}_i^c(1 - \tilde{z}_i^c) \geq w_i^2(\tilde{z}_i^c)^2$ for all i.
  - For $\tilde{z}_i^c = 1$, the constraint reduces to $\mathbf{W}_{ii} \geq w_i^2$, which is consistent with the original problem.
  - For $\tilde{z}_i^c = 0$, the constraint reduces to $\mathbf{W}_{ii} \geq 0$, which does not impose any additional restriction on w_i since $w_i \geq 0$ is already a constraint.
  - For $\tilde{z}_i^c \in (0,1)$, the constraint becomes $\mathbf{W}_{ii} \geq \frac{w_i^2}{1-\tilde{z}_i^c}$, which is a stronger constraint than the original problem, and it encourages $\tilde{z}_i^c$ to be close to 0 or 1, and discourages fractional solutions (hesitant solutions).

DPP compliance

`CvxPyLayers` [5] is a library for differentiating through convex optimization problems. It requires the problem to comply with DPP (Disciplined Parametrized Programming). As long as the problem is DPP-compliant, the library can automatically solve the optimization problem in forward passes and compute gradients in backward passes.

- For SOCP, the paper introduces artificial variables

$$\mathbf{v} = \mathbf{B}^\top \mathbf{w}, \qquad \bar{\mathbf{v}} = \Sigma_f^{1/2} \mathbf{v},$$

  and the objective can be rewritten as

$$\min \|\bar{\mathbf{v}}\|_2^2 + \operatorname{diag}(\Psi)^\top \boldsymbol{\delta}.$$

- For SDP, define

$$\bar{\mathbf{V}} := \mathbf{W}\mathbf{B}, \qquad \mathbf{V} := \mathbf{B}^\top \bar{\mathbf{V}}.$$

  Then the objective can be rewritten as

$$\min \langle \Sigma_f, \mathbf{V} \rangle + \langle \boldsymbol{\Psi}, \mathbf{W} \rangle,$$

Forward pass

For each instance j:

1. Compute factor covariance $\Sigma_f^{(j)}$.
2. Build the risk-model parameters

$$\Sigma_\theta^{(j)} = \mathbf{B}\Sigma_f^{(j)}\mathbf{B}^\top + \mathrm{diag}\Big(\exp(\hat{\psi})^2\Big)\,.$$

3. Solve the selected DPP-compliant relaxation.
4. Return relaxed portfolio weights $\mathbf{w}^{\star(j)}$.

Loss function

- After obtaining the optimal portfolio weights $\mathbf{w}^{\star(j)}$ for instance j, we can then compute the portfolio's realized return $\mathbf{r}_{\mathrm{p}(j)} = \mathbf{w}^{\star(j)\top}\mathbf{R}^{(j)} \in \mathbb{R}^T$ and evaluate its performance.
- The paper chooses to directly use the annualized Sharpe ratio:

$$\ell_d^{(j)} = \frac{(1+\mu(\mathbf{r}_{\mathrm{p}(j)}))^{365}}{\sqrt{256}\,\sigma(\mathbf{r}_{\mathrm{p}(j)})}.$$

  where $\mu(\mathbf{r}_{\mathrm{p}(j)})$ is the geometric mean, and $\sigma(\mathbf{r}_{\mathrm{p}(j)})$ is the standard deviation, with the annualization factor of 365 and 256 trading days, respectively. Further assume risk-free rate $r_f = 0$ for simplicity. Larger sharpe ratio means better performance.
- For the training batch, the loss is averaged across instances with negative sign:

$$\mathcal{L}^{(m)} = -\frac{1}{|\mathcal{B}^{(m)}|}\sum_{j\in\mathcal{B}^{(m)}} \ell_d^{(j)}.$$

Backward pass

During the backward pass, the goal is to compute the gradient $\nabla_\theta \mathcal{L}^{(m)}$ to update the risk-model parameters θ via SGD, where

$$\nabla_\theta \ell_d^{(j)} = \frac{\partial \ell_d^{(j)}}{\partial \mathbf{w}} \frac{\partial \mathbf{w}}{\partial \theta}.$$

- $\partial \ell_d / \partial \mathbf{w}$ can be explicitly computed by the chosen loss function, e.g., Sharpe ratio.
- $\partial \mathbf{w} / \partial \theta$ is the harder part, which is here automatically done by `CvxPyLayers` [5]..

Intuition of differentiating through optimization

Generally, given some optimization problem $\mathbf{w}^\star \in \arg\min P_\theta(\mathbf{w})$ as part of a neural network.

- It means that θ defines a optimization problem, and the solver returns the optimal solution $\mathbf{w}^\star$.
- Under some regularity conditions, this optimization problem is differentiable, can be backpropagated through.
- Roughly, since $\mathbf{w}^\star$ is the optimal solution, it must satisfy some optimality conditions F, e.g., KKT conditions, which contain θ and $\mathbf{w}^\star$ simultaneously. Then by implicit differentiation,

$$\frac{\partial \mathbf{w}^\star}{\partial \theta} = -\left(\frac{\partial F}{\partial \mathbf{w}^\star}\right)^{-1} \frac{\partial F}{\partial \theta}.$$

Algorithm 2: E2E learning procedure

Input: learning rate γ, schedule, batch size b, epochs
Data: factors $\mathbf{F}^{(j)}$ and returns $\mathbf{R}^{(j)}$, $j = 1, \dots, J$

```
Initialize parameters θ = {B, ψ̂}.
Distribute instances into mini-batches 𝓑^(m).
for each epoch do
    for each batch 𝓑^(m) do
        for each j ∈ 𝓑^(m) do
            compute Σ_θ^(j) using F^(j) and current θ
            solve the relaxation layer and obtain w*^(j)
            compute instance loss ℓ_d(w*^(j), R^(j))
        average batch loss 𝓛^(m) and update θ ← θ − γ∇_θ 𝓛^(m)
Use trained θ* to build the test-time covariance and solve the original sparse portfolio problem.
```

1. Initialize parameters $\boldsymbol{\theta} = \{\mathbf{B}, \hat{\psi}\}$.
2. Distribute instances into mini-batches $\mathcal{B}^{(m)}$.
3. **for** *each epoch* **do**
   - **for** *each batch* $\mathcal{B}^{(m)}$ **do**
     - **for** *each* $j \in \mathcal{B}^{(m)}$ **do**
       - compute $\Sigma_{\boldsymbol{\theta}}^{(j)}$ using $\mathbf{F}^{(j)}$ and current $\boldsymbol{\theta}$
       - solve the relaxation layer and obtain $\mathbf{w}^{\star(j)}$
       - compute instance loss $\ell_d(\mathbf{w}^{\star(j)}, \mathbf{R}^{(j)})$
     - average batch loss $\mathcal{L}^{(m)}$ and update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \gamma \nabla_{\boldsymbol{\theta}} \mathcal{L}^{(m)}$
4. Use trained $\boldsymbol{\theta}^\star$ to build the test-time covariance and solve the original sparse portfolio problem.

Experiments

Experiments

Experimental setup

Data and backtest design

- Assets: $N = 50$ S&P 500 constituents across GICS sectors.
- Factors: Fama–French five factors [2].
- Cardinality levels: $k \in \{10, 15, 20\}$.
- Date range: daily prices from 2010–2021.
- Training window: previous five years of weekly returns.
- Rebalancing: every three months; out-of-sample period 2015–2021.
- Comparison: E2E-M, E2E-SOCP, E2E-SDP, decoupled factor model (linReg), and nominal model (directly estimate covariance from data).

Circular block bootstrap [6] for training data generation i

Why more data?

- For each training period, original there is only one piece of data $\mathbf{R} \in \mathbb{R}^{T \times N}$, $\mathbf{F} \in \mathbb{R}^{T \times P}$, and is not sufficient for E2E methods to train $\boldsymbol{\theta}$.
    - For example, standing at 2015/01/01, the only available data is the weekly returns of 50 stocks and 5 factors from 2010/01/01 to 2014/12/31. Simulation is needed to generate more data for training.
- Note that during training, the data frequency is weekly. Daily data is more abundant, but it may contains more noise, more complex dynamics and instationarity.
- Also, a five-year window is also a common practice. With approximately $5 \times 52 = 260$ weekly data points, it is also a trade-off.

Circular block bootstrap [6] for training data generation ii

How to cicularly bootstrap?

- Genearlly, given a time series of length T: $X_{1:T}$, first choose a block size b and then slice the original time series into $k = T/b$ blocks, i.e.,

$$X_{1:T} = [X_{1:b}, X_{b+1:2b}, \ldots, X_{(k-1)b+1:kb}] = [B_1, B_2, \ldots, B_k]$$

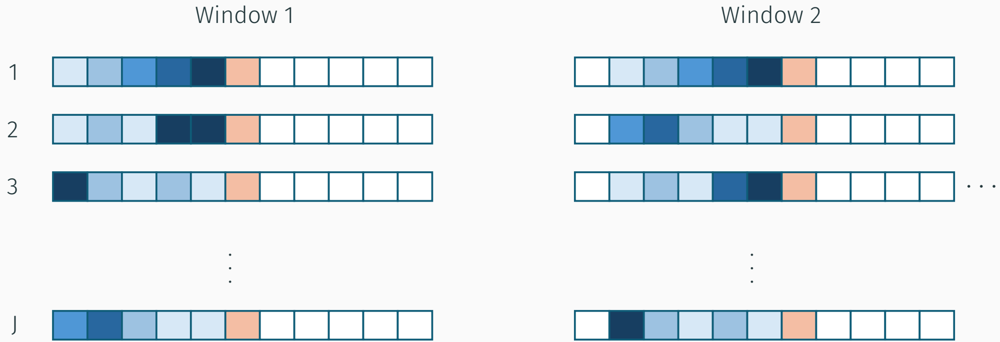
- Then for bootstrapping, randomly sample k blocks with replacement from $\{B_1, \ldots, B_k\}$ and concatenate them to get a bootstrapped time series of length T:

$$\tilde{X}_{1:T}^{(j)} = [\tilde{B}_1^{(j)}, \tilde{B}_2^{(j)}, \ldots, \tilde{B}_k^{(j)}].$$

- The circular block bootstrap preserves the temporal dependence structure within each block, and by sampling blocks with replacement, it can generate new time series that are similar to the original one but with some randomness, which is useful for training the E2E model.
- In the paper, the block size is chosen as $b = 20$ weeks, and the paper generates $J = 2000$ bootstrapped time series for training.

Detailed training procedure

- For each training, a five-year window of weekly returns is used to train the E2E model (e.g., the first training period is 2010Q1 $\sim$ 2014Q4). With such many data (and by bootstrapping), the paper will compute the factor covariance $\Sigma_f^{(j)}$ for each instance j and then train the E2E model by SGD to get the optimal parameters $\theta^\star$.

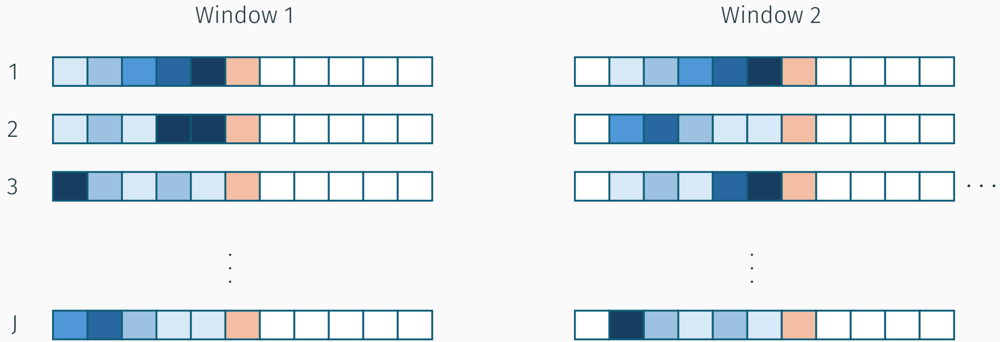


Detailed training procedure

- After training, the paper will use the trained $\theta^\star$ to build the test-time covariance $\Sigma_{\theta^\star}$ and solve the original sparse portfolio problem

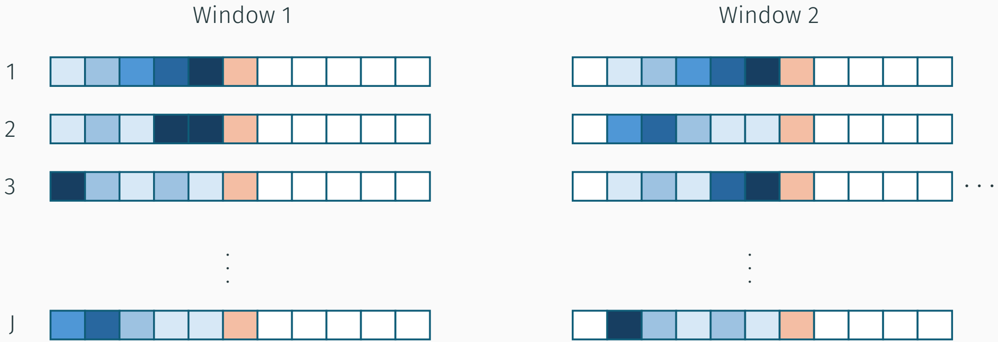
$$\min_{\mathbf{w}\in\mathcal{W}_k} \mathbf{w}^\top \Sigma_{\theta^\star} \mathbf{w}.$$

directly by Gurobi to get the final portfolio weights $\mathbf{w}^\star$ for out-of-sample evaluation.



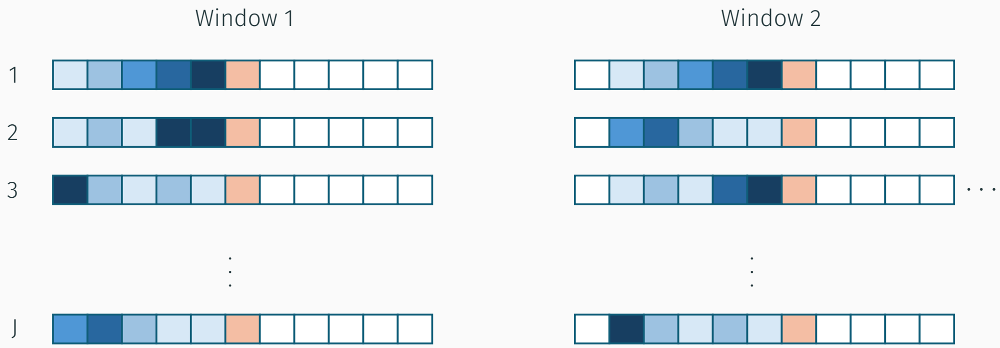
Detailed training procedure

- During evaluation, the model will buy-and-hold the portfolio for three months to get the realized return $\mathbf{r}_{\mathrm{p}} \in \mathbb{R}^T$ and compute the performance metrics, e.g., annualized return, volatility, Sharpe ratio, max drawdown, etc.



Detailed training procedure

- This procedure is repeated for every three months from 2015 to 2021 (so the next training period is 2010Q2 $\sim$ 2015Q1 and evaluate on 2015Q2), and the paper will report the average performance across all these out-of-sample periods.

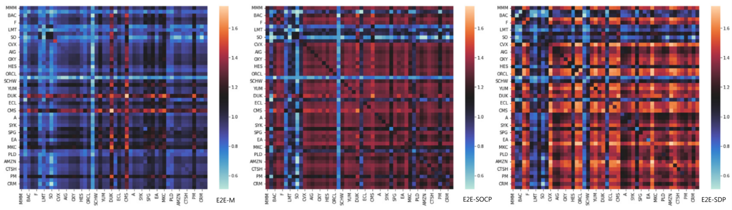


Experiments

Results

Training behavior

1. Loss usually drops sharply in the first epoch (Prcatically, the paper initializes θ from the decoupled fit as a warm start).
2. E2E training materially changes the covariance matrix relative to the decoupled fit.



3. Training cost is not neglectable.

Layer	Structure	Reported epoch-time scale
E2E M	convex QP	about 15 seconds
E2E SOCP	conic perspective constraints	about 1 minute
E2E SDP	lifted semidefinite cone	over 6 hours

In-sample result: E2E beats linReg

Across 24 training periods, the paper counts pairwise wins.

Metric against linReg	$k = 10$	$k = 15$	$k = 20$
E2E M variance wins	$21/24$	$19/24$	$19/24$
E2E SOCP variance wins	$23/24$	$21/24$	$20/24$
E2E SDP variance wins	$21/24$	$19/24$	$19/24$
Any E2E return wins	$24/24$	$24/24$	$24/24$
Any E2E Sharpe wins	$24/24$	$24/24$	$24/24$

- The integrated learner improves in-sample returns and Sharpe ratios consistently, with little or no volatility penalty.

Appendix: out-of-sample metrics

Metric	Interpretation
Annualized return	geometric average return scaled to one year
Annualized volatility	standard deviation of portfolio returns scaled by $\sqrt{256}$
VaR / CVaR	tail-loss measures at $\alpha = 0.05$
LPM2	second-order lower partial moment below zero return
Max drawdown	largest peak-to-trough wealth decline
Sharpe / Sortino / Calmar	risk-adjusted returns using volatility, downside risk, or drawdown
Information ratio	active return over tracking error relative to nominal benchmark

Out-of-sample risk and return

Metric	k	E2E M	E2E SOCP	E2E SDP	linReg	nominal
Ann. ret.	10	.1908	.1776	.1699	.1535	.1684
	15	.1707	.1697	.1698	.1424	.1533
	20	.1729	.1687	.1708	.1441	.1550
Ann. vol.	10	.1674	.1695	.1697	.1755	.1686
	15	.1686	.1667	.1670	.1701	.1680
	20	.1687	.1669	.1682	.1700	.1674
Max DD	10	.3149	.3361	.3331	.3585	.3417
	15	.3358	.3347	.3208	.3491	.3657
	20	.3357	.3348	.3370	.3476	.3632

- Every E2E variant has higher annualized return than linReg at all cardinalities.
- E2E also reduces annualized volatility relative to linReg.
- Compared with nominal, E2E has better return, volatility and maximum drawdown; however, nominal has better VaR and CVaR.
- Inside the E2E variants, Big-M has the best return and Sharpe, with also higher volatility.

Out-of-sample risk-adjusted metrics

Metric	k	E2E M	E2E SOCP	E2E SDP	linReg	nominal
Sharpe	10	1.1399	1.0014	1.0474	.8747	.9993
	15	1.0126	1.0169	1.0180	.8370	.9128
	20	1.0247	1.0155	1.0106	.8478	.9265
Sortino	10	2.3407	2.0179	2.1009	1.6560	2.0171
	15	2.0391	2.0677	2.0577	1.6301	1.7960
	20	2.0606	2.0413	2.0420	1.6602	1.8397
Calmar	10	.6059	.5102	.5284	.4280	.4930
	15	.5083	.5294	.5071	.4079	.4193
	20	.5150	.5068	.5039	.4145	.4269

- E2E is consistently significantly better than linReg, and also better than nominal.
- Compared with nominal, E2E brings positive extra return adjusted by risk, while linReg does the opposite.

Discussion

Counterintuitive relaxation performance

The original intuition from combinatorial optimization:

$$\text{tighter relaxation} \quad \Rightarrow \quad \text{better surrogate} \quad \Rightarrow \quad \text{better decisions.}$$

The paper's evidence is more complicated:

- E2E M often posts the strongest out-of-sample risk-adjusted performance.
- SOCP and SDP can reduce volatility, but do not clearly dominate in returns.
- The tighter layers are far more expensive and have larger optimization architectures.

References

[1] Adam N. Elmachtoub and Paul Grigas. **"Smart "Predict, then Optimize""**. In: *Management Science* 68.1 (2021), pp. 9–26.

[2] Eugene F. Fama and Kenneth R. French. **"A Five-factor Asset Pricing Model"**. In: *Journal of Financial Economics* 116.1 (2015), pp. 1–22.

[3] Xiangyu Cui, Xiaoling Zheng, Shushang Zhu, and Xiaoling Sun. **"Convex Relaxations and MIQCQP Reformulations for a Class of Cardinality-constrained Portfolio Selection Problems"**. In: *Journal of Global Optimization* 56.4 (2013), pp. 1409–1423.

[4] Angelika Wiegele and Shun Zhao. **"Tight SDP Relaxations for Cardinality-constrained Problems"**. In: *Operations Research Proceedings 2021*. Springer, 2022, pp. 167–172.

[5] Akshay Agrawal, Brandon Amos, Shane Barratt, Stephen Boyd, Steven Diamond, and J. Zico Kolter. **"Differentiable Convex Optimization Layers"**. In: *Advances in Neural Information Processing Systems*. Vol. 32. 2019, pp. 9562–9574.

[6] Dimitris N. Politis and Joseph P. Romano. **"A Circular Block-resampling Procedure for Stationary Data"**. In: *Exploring the Limits of Bootstrap*. Wiley, 1991, pp. 263–270.